

PORTFOLIO PROJECT

Netflix Data Analysis using Python

Exploratory Data Analysis (EDA)

8,807 titles | 12 columns | Movies and TV Shows



Python EDA

A catalog story built from data

Objective

Use Python-based exploratory analysis to understand Netflix catalog composition, release patterns, geographic contribution, genres, ratings, and talent concentration.



Business questions answered

What dominates the library? Where does content come from? Which genres, ratings, durations, release years, months, and creators stand out?



TITLES

8,807

Real-world Netflix rows



COLUMNS

12

Metadata dimensions



MOVIES

6,131

69.6% of catalog



TV SHOWS

2,676

30.4% of catalog



What the data contains

8,807

Netflix titles analyzed

A real-world catalog dataset covering title metadata, release year, date added, audience rating, duration, genre tags, country, cast, and director.



12 Columns

A compact schema covering content identity, people, timing, ratings, duration, country, and genre.



Movies and TV Shows

The dataset supports direct comparison between long-form films and episodic series.



Catalog Metadata

The analysis uses counts, crosstabs, histograms, rankings, and time-based grouping.



Portfolio Ready

The project demonstrates cleaning, EDA, visualization, and business insight generation.



From raw catalog to analysis-ready data



Missing values

Filled director, cast, country, and rating gaps with Unknown.



Date text cleaned

Trimmed spacing in the date_added column.



Datetime parsing

Converted date_added into datetime for month analysis.



Categories prepared

Kept type, rating, country, and listed_in ready for grouping.



EDA-ready table

Exported the cleaned CSV and generated Matplotlib charts.

Cleaning choice to note: Unknown remains visible in the country chart, making missing geography transparent instead of silently dropping those titles.

Technical stack behind the EDA

Technologies



Python

Core language for loading, cleaning, grouping, and transforming the dataset.



Pandas

Used for missing-value handling, date conversion, counts, crosstabs, and ranking.



Matplotlib

Generated every exported chart PDF used in this presentation.

Analyst capabilities



Data Cleaning

Prepared missing values, dates, and categorical fields.



EDA

Explored distributions, rankings, trends, and splits.



Visualization

Turned grouped data into readable charts.



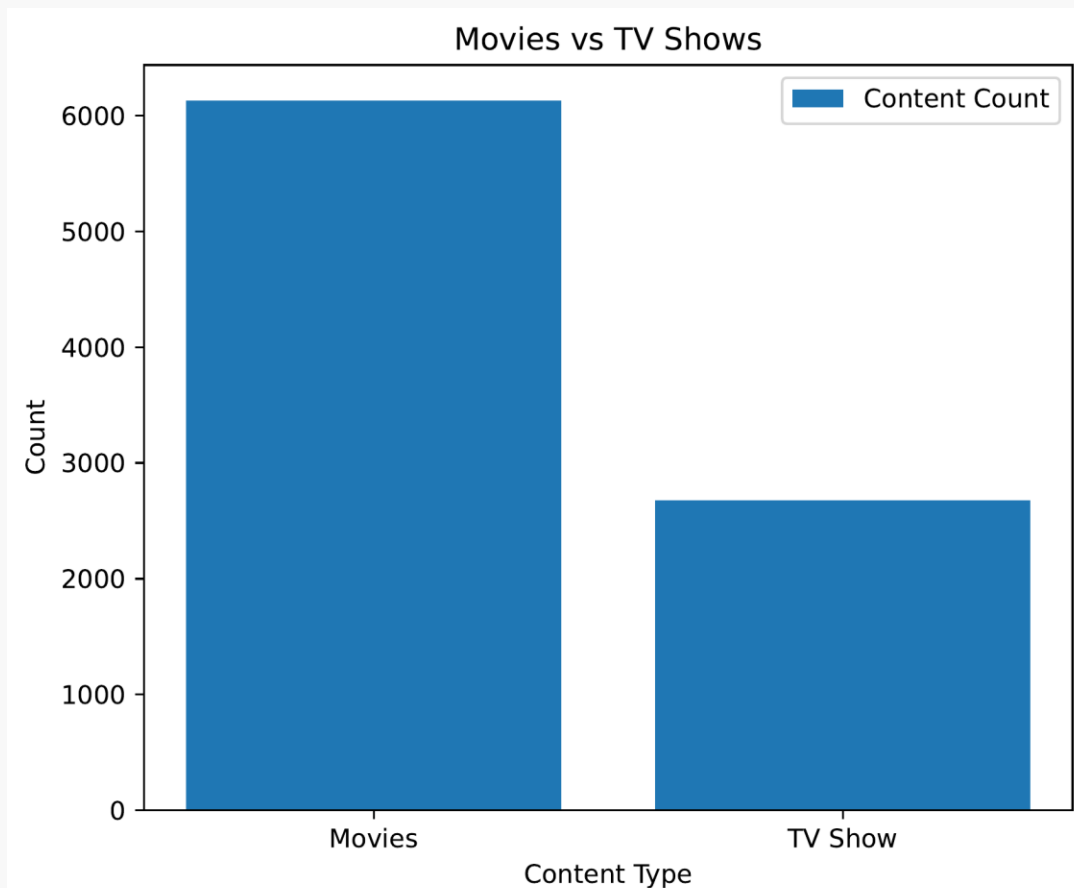
Insight Generation

Translated charts into recruiter-ready business takeaways.

Movies vs TV Shows



06



BUSINESS QUESTION

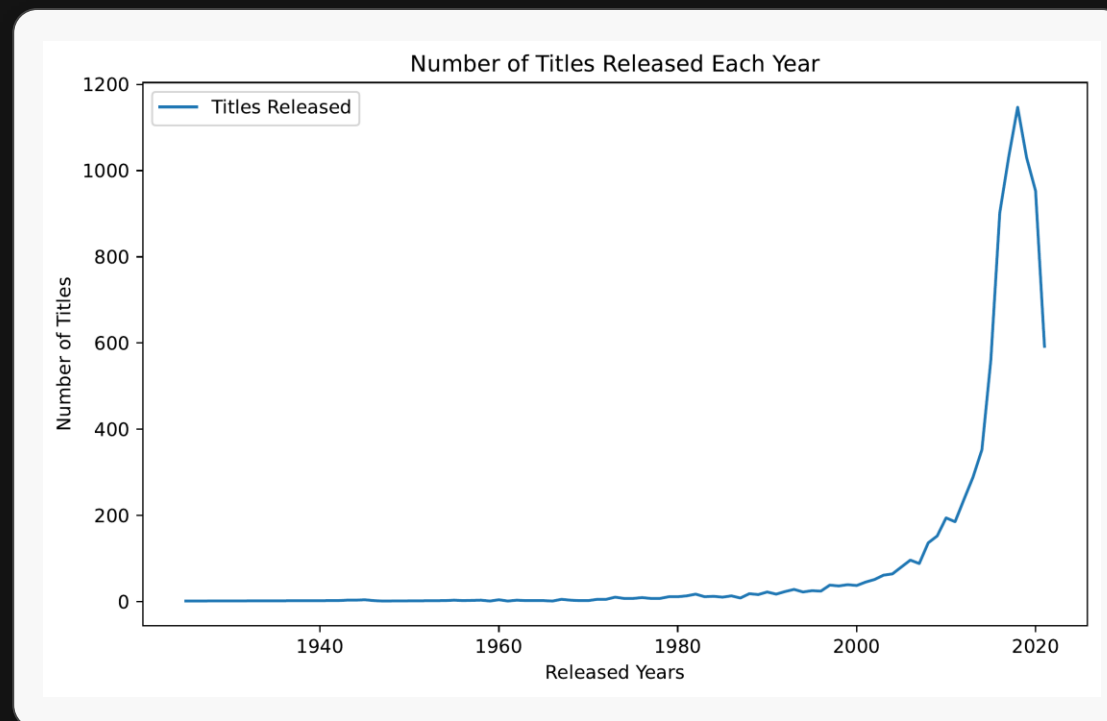
What type of content dominates Netflix?

KEY INSIGHT

Movies lead the catalog with 6,131 titles (69.6%).

TV Shows account for 2,676 titles, showing a catalog that is still film-heavy but broad enough for episodic content.

Number of Titles Released Each Year



BUSINESS QUESTION

How has content production changed over time?



KEY INSIGHT

Release volume peaks in 2018 with 1,147 titles.



WHY IT MATTERS

The trend rises sharply in the modern streaming era, highlighting how strongly the catalog is weighted toward recent production.



Top Countries Producing Netflix Content



BUSINESS QUESTION

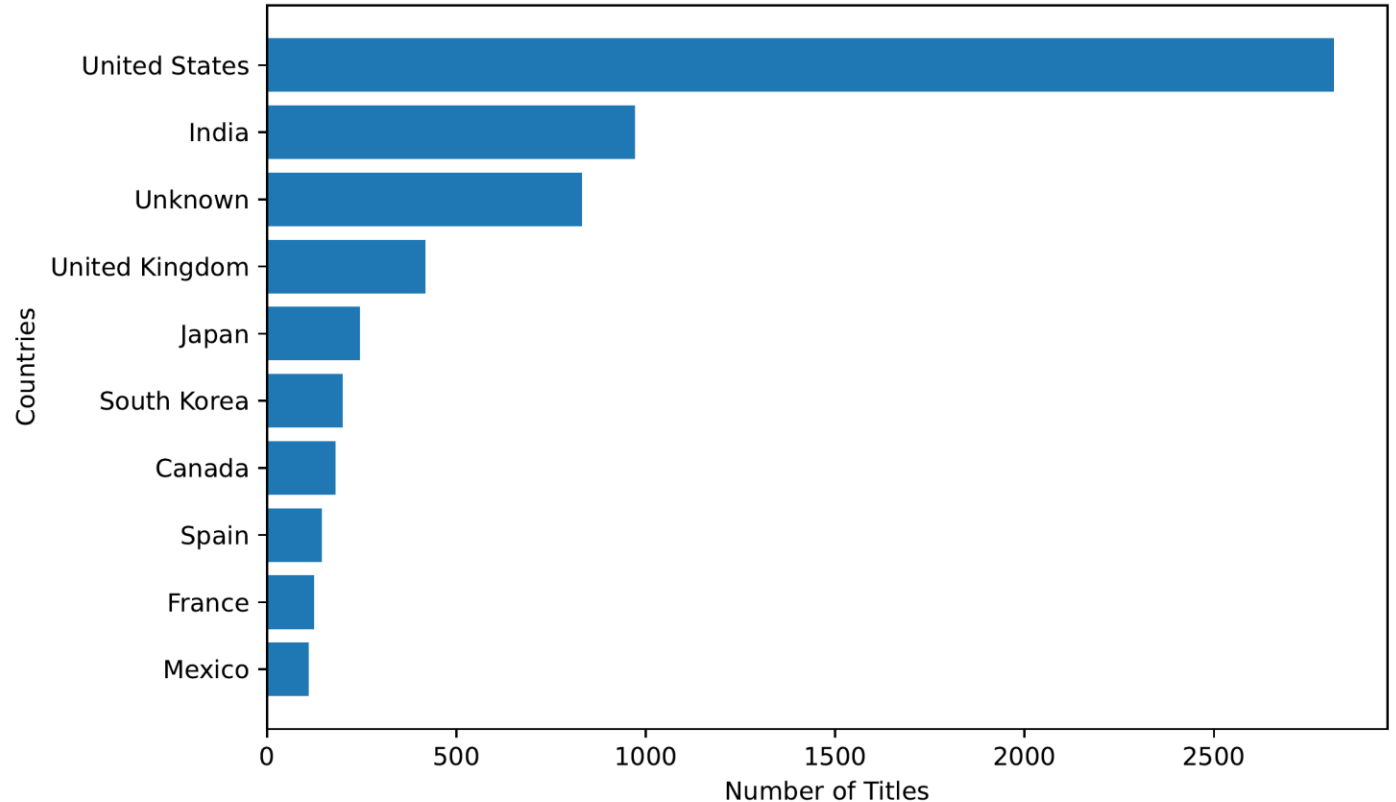
Which countries contribute the most titles?

KEY INSIGHT

The United States leads with 2,818 titles; India follows with 972.

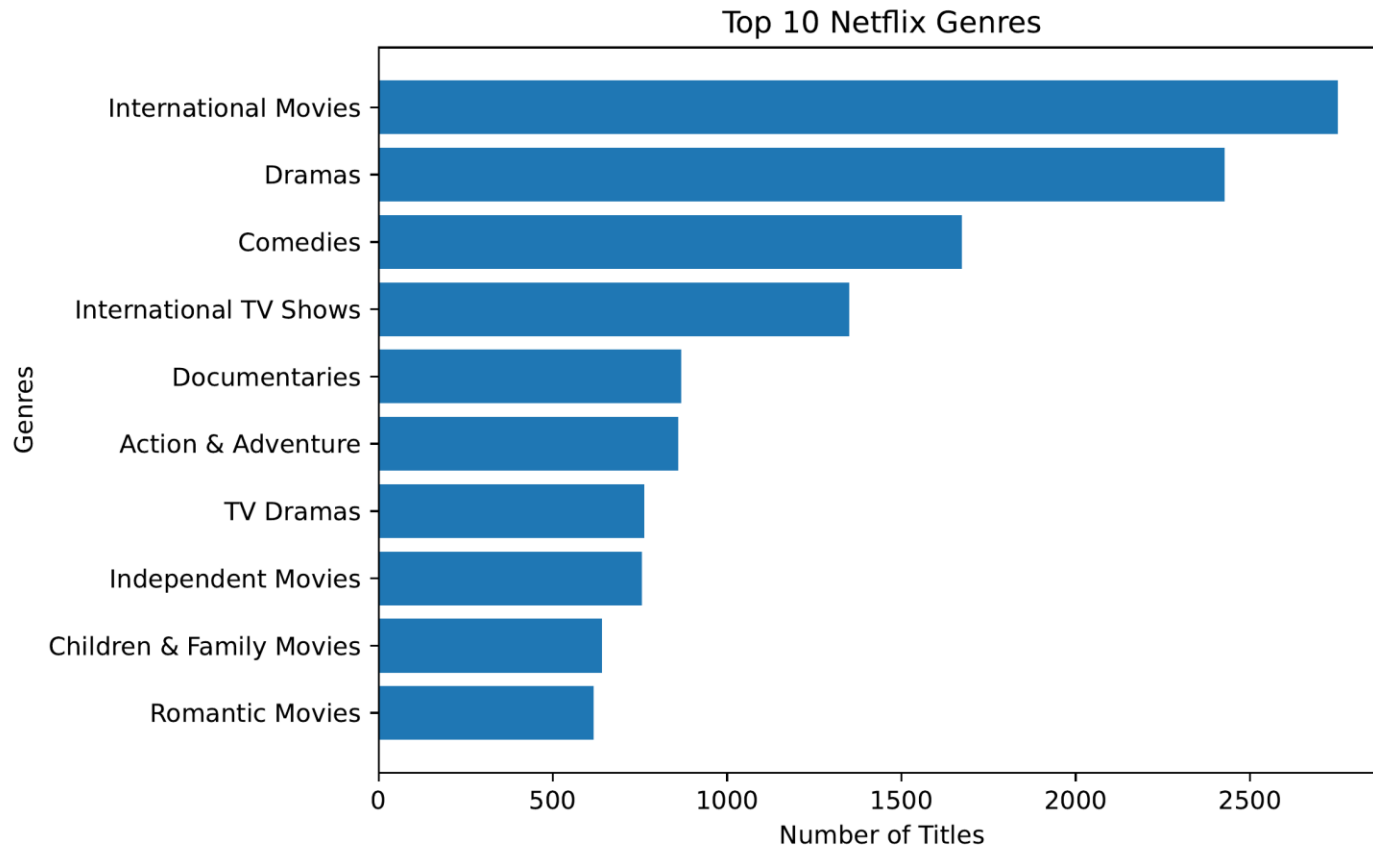
Unknown appears as a visible category, which reflects the cleaning decision to preserve missing geography as an auditable bucket.

Top 10 Countries Producing Netflix Content





Top 10 Netflix Genres



BUSINESS QUESTION

What genres does Netflix focus on most?

KEY INSIGHT

International Movies ranks first with 2,752 titles.

Dramas and Comedies also rank highly, showing a catalog built around globally scalable, broad-audience categories.

Content Ratings Distribution



BUSINESS QUESTION

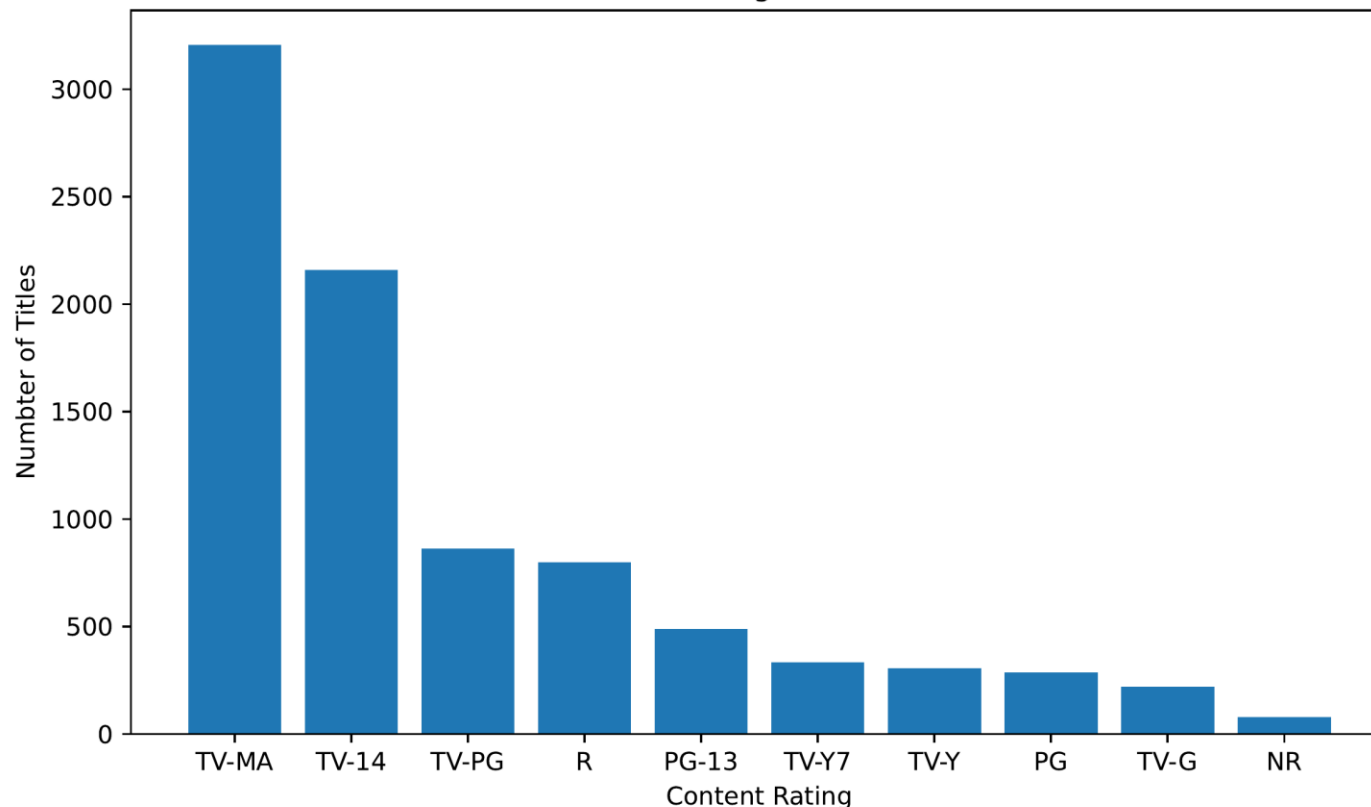
Is the catalog family-friendly or adult-oriented?

KEY INSIGHT

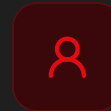
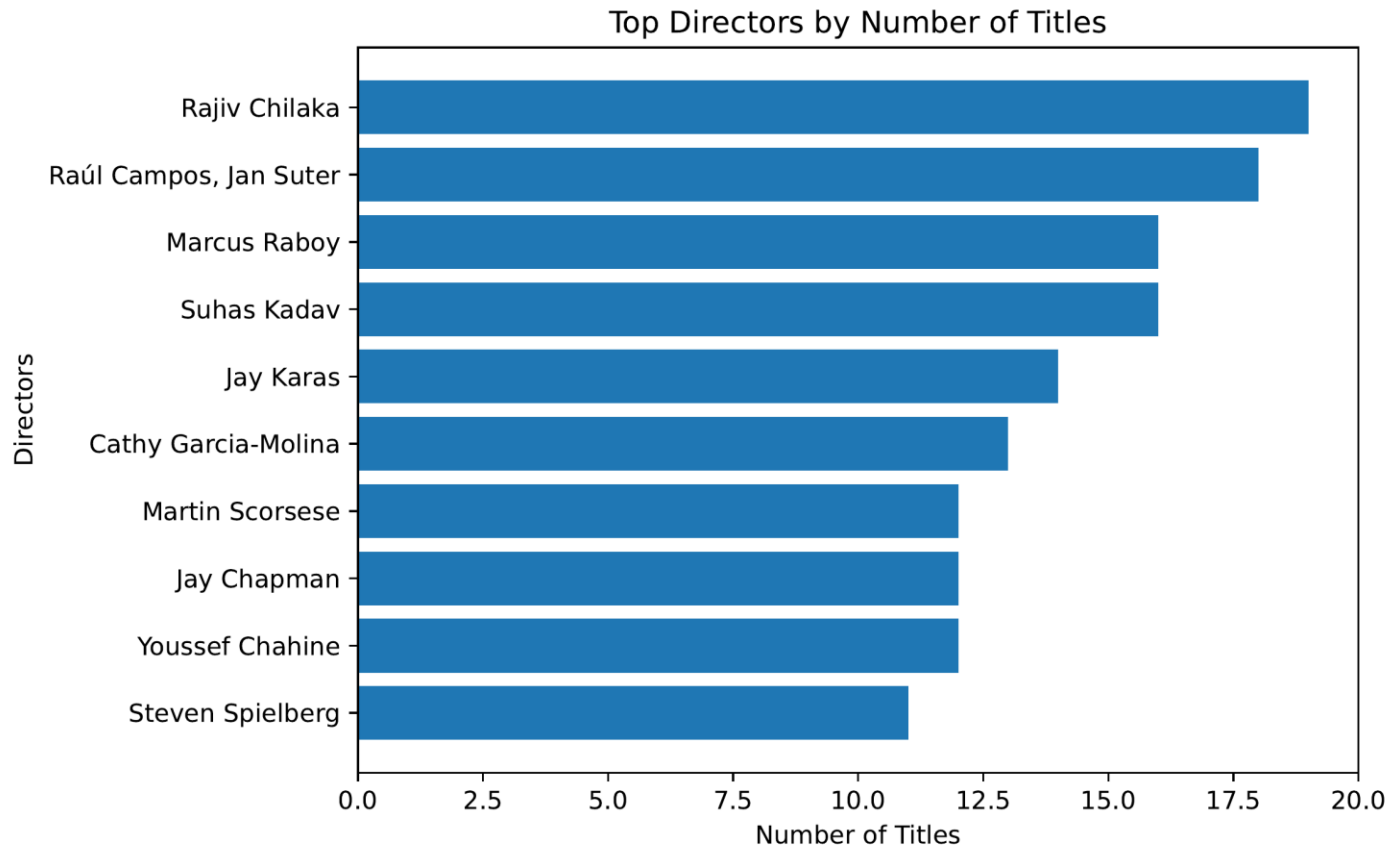
TV-MA is the most common rating with 3,207 titles (36.4%).

The top two ratings, TV-MA and TV-14, suggest the catalog leans toward mature and teen audiences.

Content Ratings Distribution



Top Directors by Number of Titles



BUSINESS QUESTION

Which directors appear most frequently?

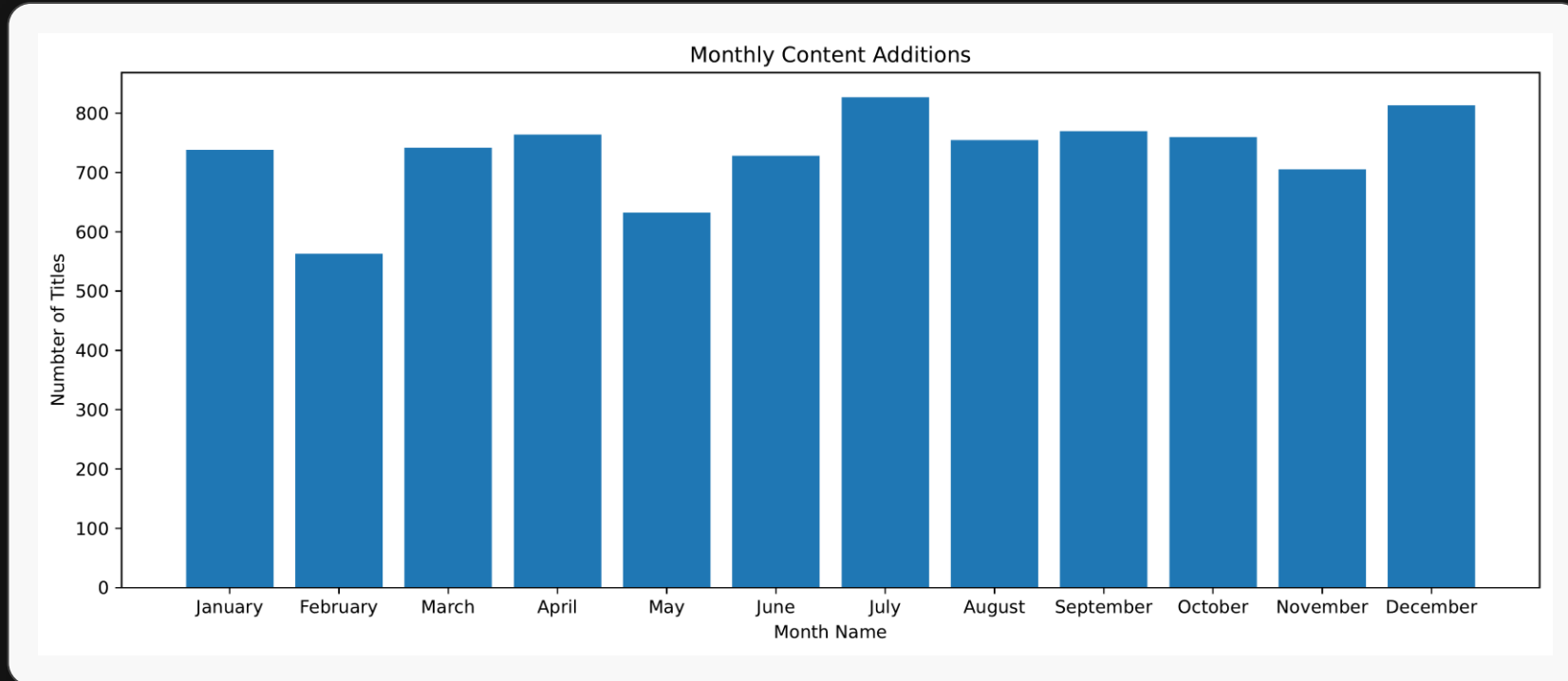
KEY INSIGHT

Rajiv Chilaka leads the director list with 19 titles.

The top director counts are relatively close, suggesting creator concentration exists but is not dominated by a single name.



Monthly Content Additions



BUSINESS QUESTION

During which months does Netflix add the most content?



KEY INSIGHT

July leads with 827 additions.



WHY IT MATTERS

Monthly additions vary across the year, revealing release timing patterns that can support programming and campaign planning.

Movie Duration Distribution



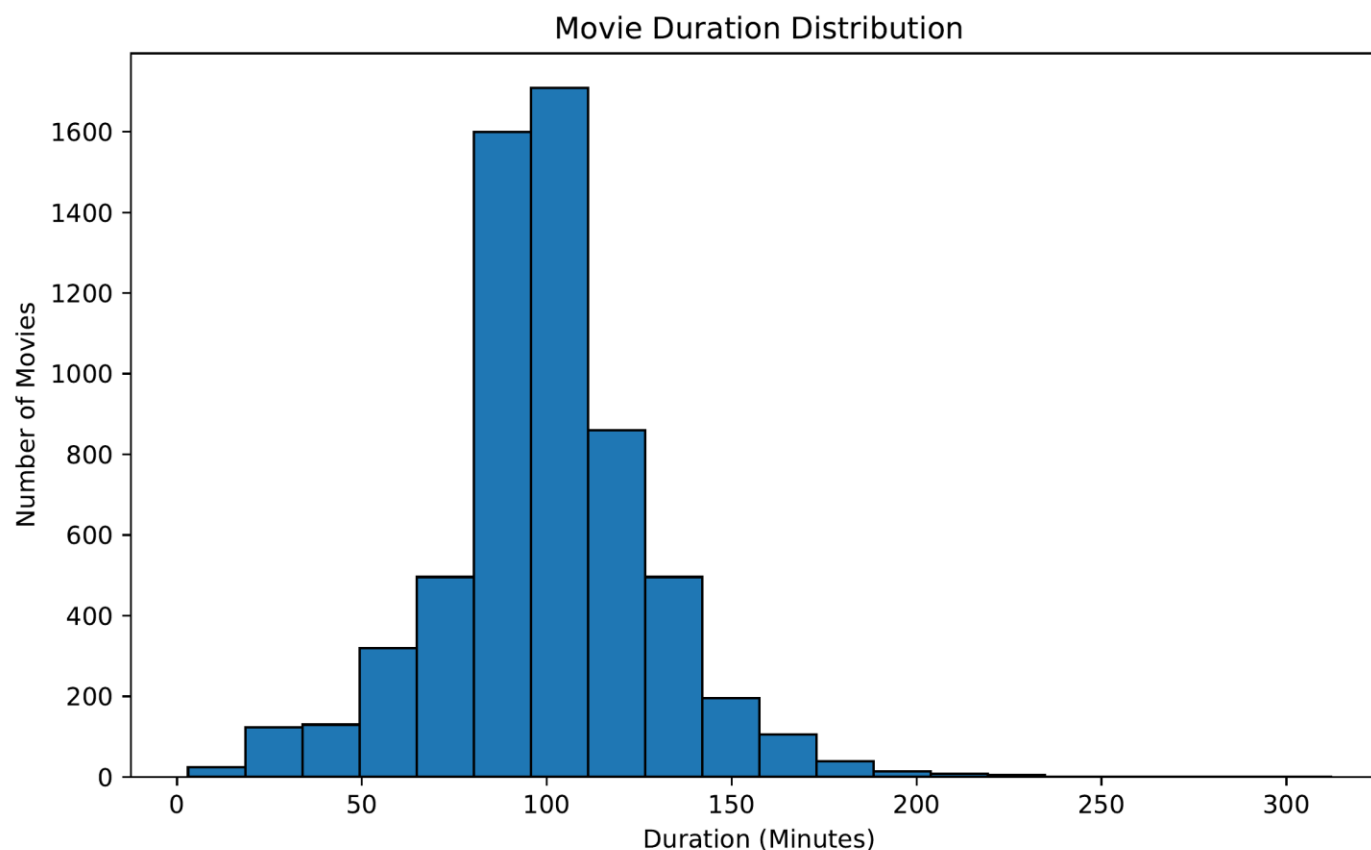
BUSINESS QUESTION

What is the typical movie length?

KEY INSIGHT

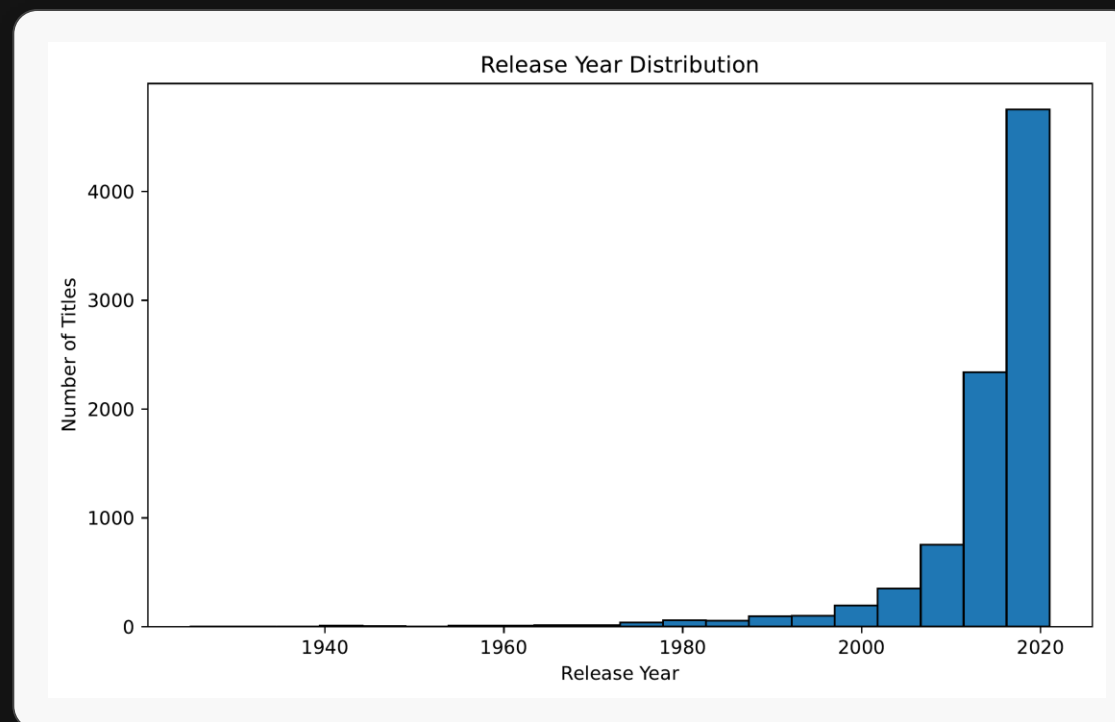
Median movie duration is 98 min.

51.4% of movies fall between 90 and 120 minutes, matching the common feature-film range.





Release Year Distribution



BUSINESS QUESTION

Does Netflix feature newer releases or older content?



KEY INSIGHT

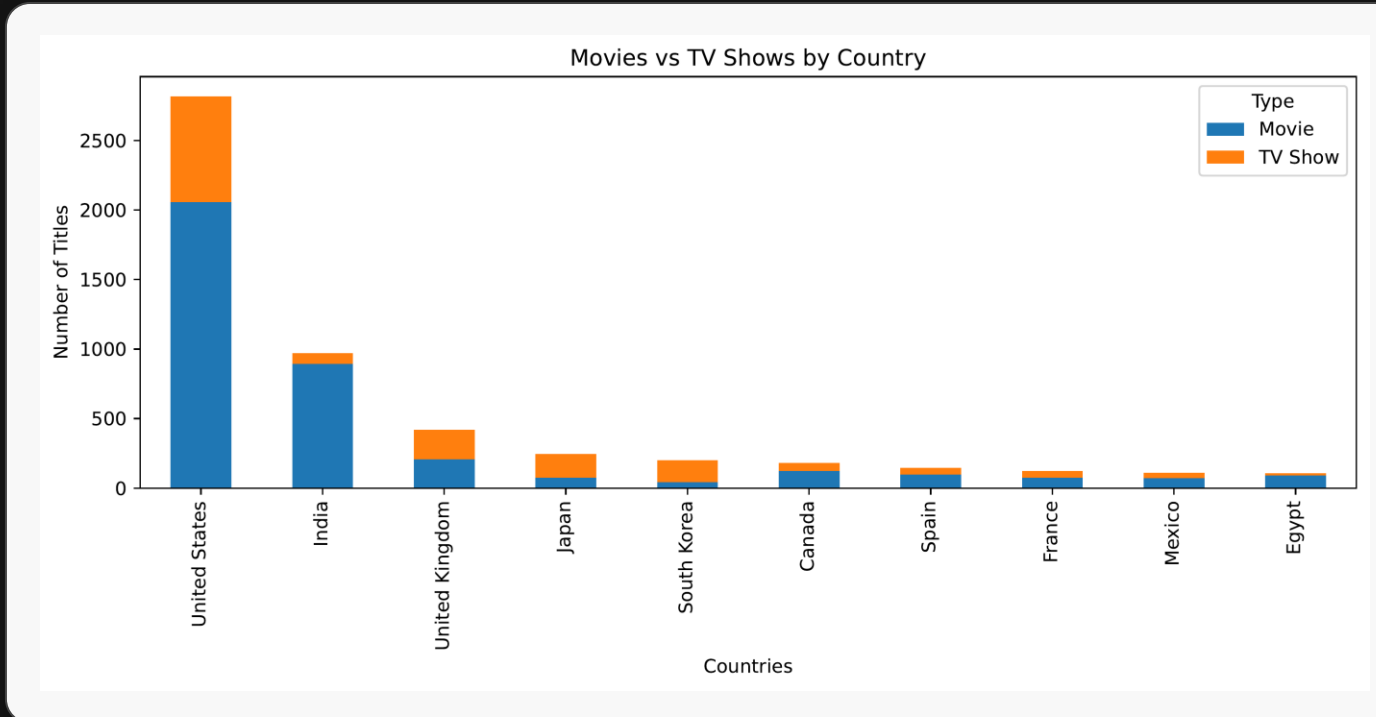
84.8% of titles are from 2010 or later.



WHY IT MATTERS

The distribution is strongly weighted toward recent years, which is consistent with a streaming catalog built around modern availability.

Movies vs TV Shows by Country



BUSINESS QUESTION

Which countries produce more movies versus TV shows?



KEY INSIGHT

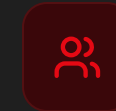
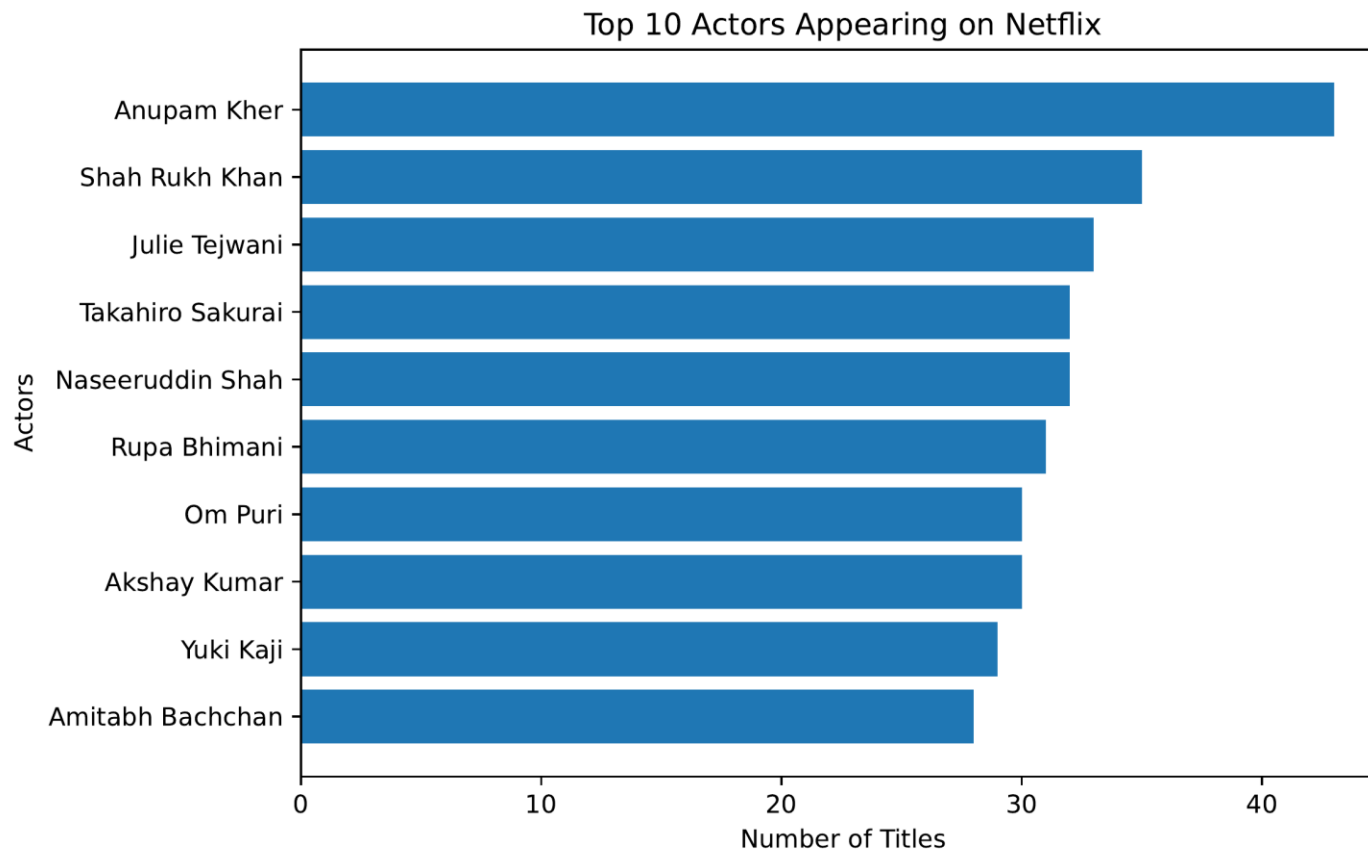
The United States and India are movie-heavy; Japan and South Korea skew toward TV shows.



WHY IT MATTERS

Format mix differs by market, which can shape regional licensing, localization, and audience strategy.

Top 10 Actors Appearing on Netflix



BUSINESS QUESTION

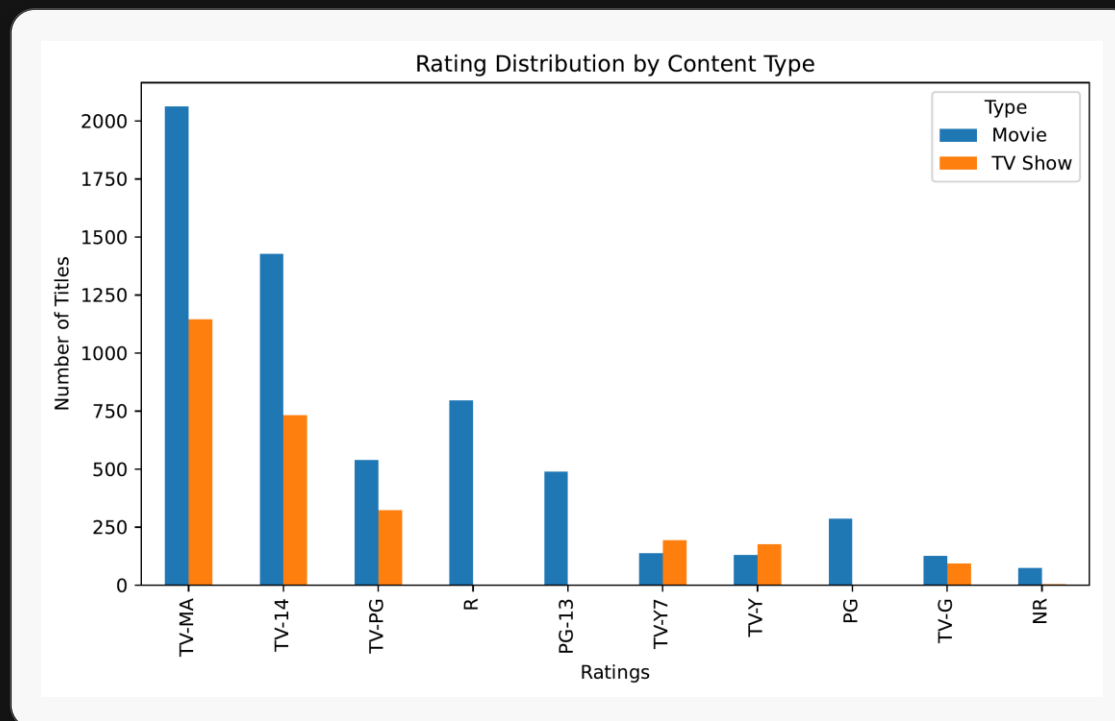
Which actors are featured most often?

KEY INSIGHT

Anupam Kher appears most often with 43 titles.

Several high-ranking actors are associated with Indian cinema, reinforcing the catalog depth represented by India in the country analysis.

Rating Distribution by Content Type



BUSINESS QUESTION

Do movies and TV shows target different audiences?



KEY INSIGHT

TV-MA and TV-14 dominate both formats; R and PG-13 are almost entirely movies.



WHY IT MATTERS

The rating split clarifies how audience targeting differs between films and series inside the catalog.



What the Netflix catalog analysis reveals

Movies dominate the library

6,131 movies represent 69.6% of the catalog.

Adult-oriented ratings are common

TV-MA is the top rating with 3,207 titles (36.4%).

Release volume surged after 2015

2018 is the peak release year with 1,147 titles.

Movies cluster near standard feature length

Median duration is 98 min; 51.4% fall between 90 and 120 minutes.

The United States contributes the most

2,818 titles are listed under the United States.

Release timing has seasonal patterns

July leads monthly additions with 827 titles.

International content is a major pillar

International Movies leads genres with 2,752 titles.

THANK YOU

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Netflix EDA